Card 1 Cold-Start Thermal Stress, 2-Beat Analog Bit Slicing & Integrated-L2 Anti-Windup

2x2 Factorial Validation, Multi-Round Hyperparameter Optimization & Forward Hadamard Propagation (FHP)

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Executive Summary & Scientific Scope: This report validates closed-loop Mode A 2-beat analog bit slicing (**K=1**, **40%** heater PWM, **1k/2.2k** attenuator) after resolving two-layer backpropagation sign, **M3** scaling (>>6) and **±25 LSB** active-row capping, Ethernet frame length acceptance, and integrated-L2 back-calculation anti-windup (**T120_SOC_ETH_FW v2.2-MAML**). Unheated cold verification achieved **9.45 b** (ON) and **9.45 b** (OFF, W8=0). Primary highlighted run (**opt_r2_lr0p035_l2_1536_mom0p85_cl75.json**, 4,013 pairs: **LR=0.035, L2=1536, Mom=0.85, Clamp=±75 LSB**) achieved detrended Warm-ON lift **+0.50 b**, loss ratio **1.38x**, climb **+0.82 b**, and anti-windup recovery **0.96** (Score **+2.004**).

1. 2x2 Factorial Performance Accounting & Latest Hyperparameter Optimization

Factorial Condition	Thermal State	Regulation	Mean Loss	Mean ENOB	Peak ENOB	W8 Env [MSB / LSB]
1. Cold Closed-Loop	Heaters OFF	ON (K=1)	3,694 L²	**9.43 b**	9.50 b	[-14, 13] / [-10, 9]
2. Cold Open-Loop	Heaters OFF	OFF (W8=0)	3,803 L²	**9.43 b**	9.50 b	[0,0] (0% leak)
3. Warm Closed-Loop	Heaters ON 40%	ON (K=1)	**16,811 L²**	**8.69 b**	9.50 b	[-43, 75] / [-44, 75]
4. Warm Open-Loop	Heaters ON 40%	OFF (W8=0)	32,802 L²	7.89 b	9.50 b	[0,0] (0% leak)

Latest Hyperparameter & Detrended Optimization Results (R1–R3)

Rnd	LR	L2 Reg	Mom	Clamp	Lift vs E_base	Loss Ratio	Hot Climb	Jitter	Recov	Score
R1	0.03	1024	0.9	±75	**+0.66 b**	1.57x	+0.74 b	0.33 b	0.67	**+1.957**
R2 ★	0.035	1536	0.85	±75	**+0.50 b**	1.38x	+0.82 b	0.29 b	0.96	**+2.004**
R3	0.02	2560	0.8	±70	**+0.09 b**	1.11x	+0.22 b	0.26 b	1.00	**+0.559**

Next Active GP Model Target (Round 4): LR=0.045, L2=2048, Momentum=0.88, Clamp=±75 LSB (predicted Score +1.830 ± 0.822).

2. Cold-Start Calibration, 2-Beat Bit Slicing & Phase-by-Phase Telemetry

• **Cold-Start 60s Cooldown & Single-Shot DC Calibration:** Heaters are held at 0% for 60 seconds before each experiment so the Trion T120 SoC locks static (alpha, beta) DC calibration vectors on cold silicon (0x1E subcmd 5). Zero surrogate re-calibrations occur at runtime.
• **Integrated-L2 Back-Calculation Anti-Windup:** Under heavy thermal stress, weights hitting ±75 LSB bleed 1/4 of their overshoot back out of the momentum integrator, allowing instantaneous release when error gradients reverse without overshoot lag. Matrix 3 active rows 0–5 are capped to ±25 LSB so rows 6–7 clock staircases (±40 LSB) never shift.

Phase-by-Phase Telemetry (opt_r2_lr0p035_l2_1536_mom0p85_cl75.json, 4,013 Pairs)

Ph	Regime	Pairs	Comb Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	331	9.41->9.44 (**9.43b**)	—	3743->3608	[14,10]	0	330
P2	Cold / Regler OFF	309	9.45->9.45 (**9.43b**)	—	3180->3834	[0,0]	0	0
P3	Warm / Regler ON	1084	9.45->8.96 (**9.23b**)	**+0.12 b**	3702->10297	[51,39]	4	1083
P4	Warm / Regler OFF	952	8.92->8.03 (**8.31b**)	**-0.03 b**	12876->29416	[0,0]	0	0
P5	Warm / Regler ON	850	7.93->7.93 (**8.00b**)	**+0.72 b**	31966->31202	[75,75]	16	850
P6	Warm / Regler OFF	487	7.11->6.90 (**7.06b**)	**+0.03 b**	49599->55672	[0,0]	0	0

3. Empirical Visualizations: Full Real-Time Timeline

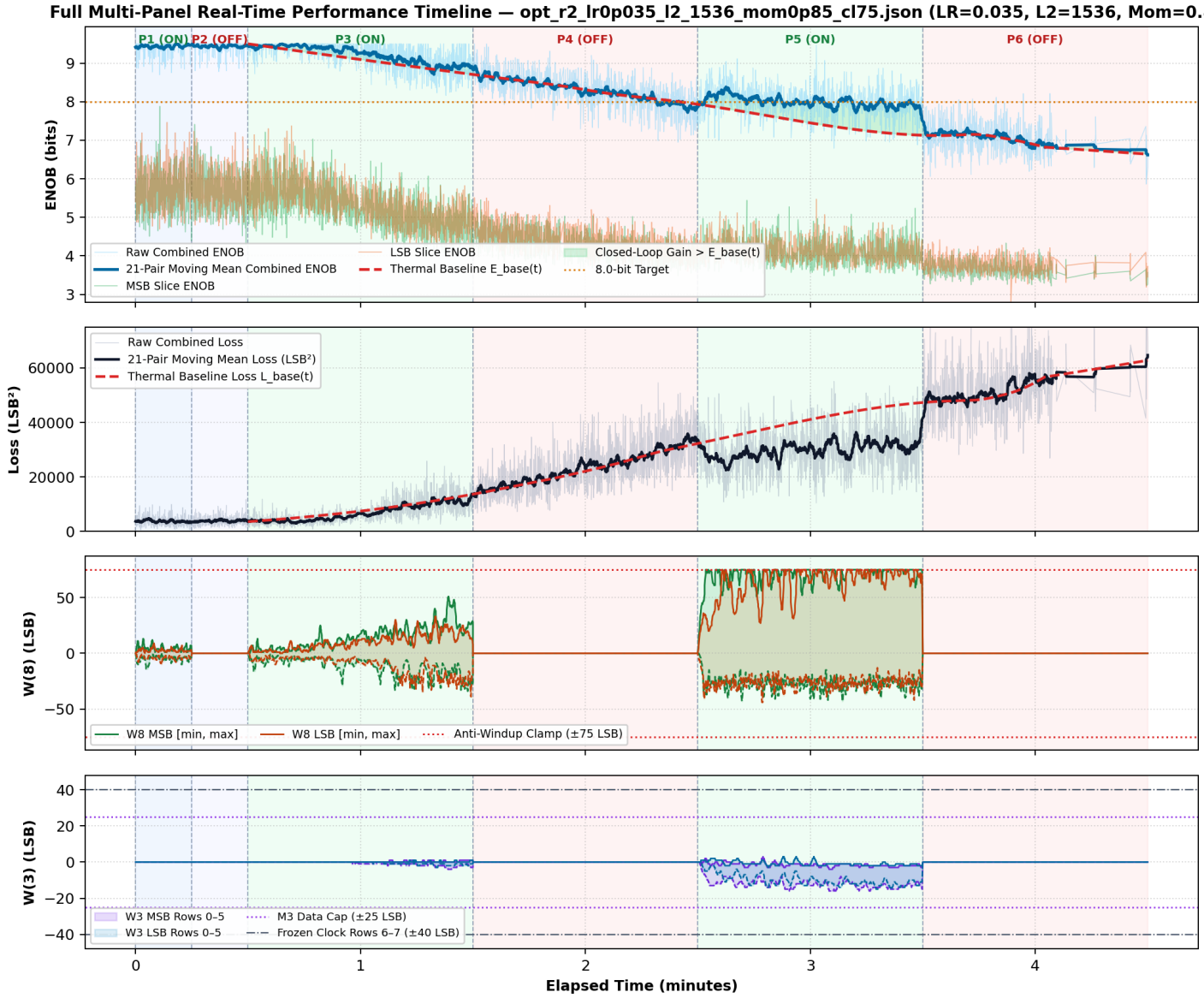


Figure 1: Full Multi-Panel Real-Time Performance Timeline. (Panel 1) Combined 8-bit ENOB (blue) alongside individual MSB (green) and LSB (orange) slices, overlaid with the continuous PCHIP uncompensated thermal drift baseline E_base(t) (red dashed) interpolated across Cold-OFF and Warm-OFF phases. (Panel 2) Combined hardware MSE loss vs L_base(t). (Panel 3) Matrix 8 MSB & LSB weight envelopes bounded by integrated-L2 anti-windup rails (±75 LSB). (Panel 4) Matrix 3 active data rows 0–5 (±25 LSB cap) keeping reference clock staircases (±40 LSB) undisturbed.

Multi-Round & Phase-by-Phase Thermal Drift Heatmap Comparison

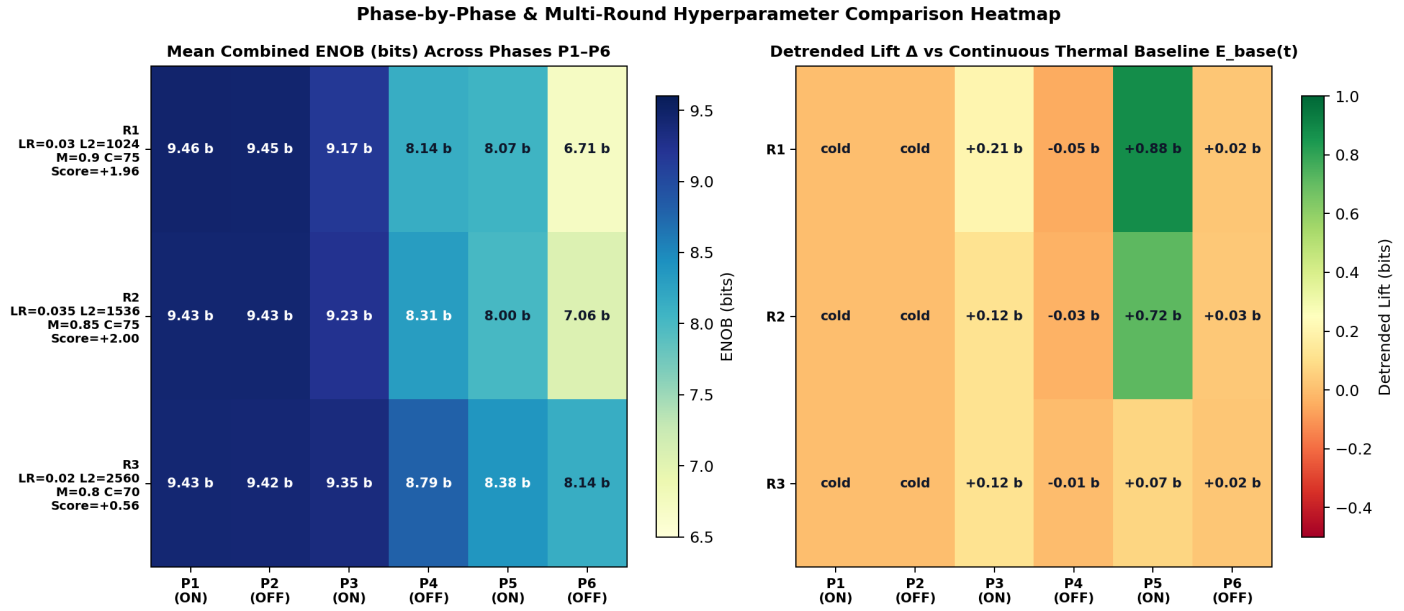


Figure 2: Phase-by-Phase ENOB & Detrended Lift Heatmap Across Rounds R1–R3. Left compares mean Combined ENOB across all 6 phases; Right shows the detrended closed-loop lift (ΔENOB vs continuous PCHIP baseline E_base(t)) during Warm phases P3–P6, demonstrating deterministic collapse near 0.00 b whenever Mode A is turned OFF and sharp closed-loop recovery when Mode A engages.

Matrix 8 Full Synaptic Telemetry: All 84 W(8) Weights (7x6 MSB + 7x6 LSB)

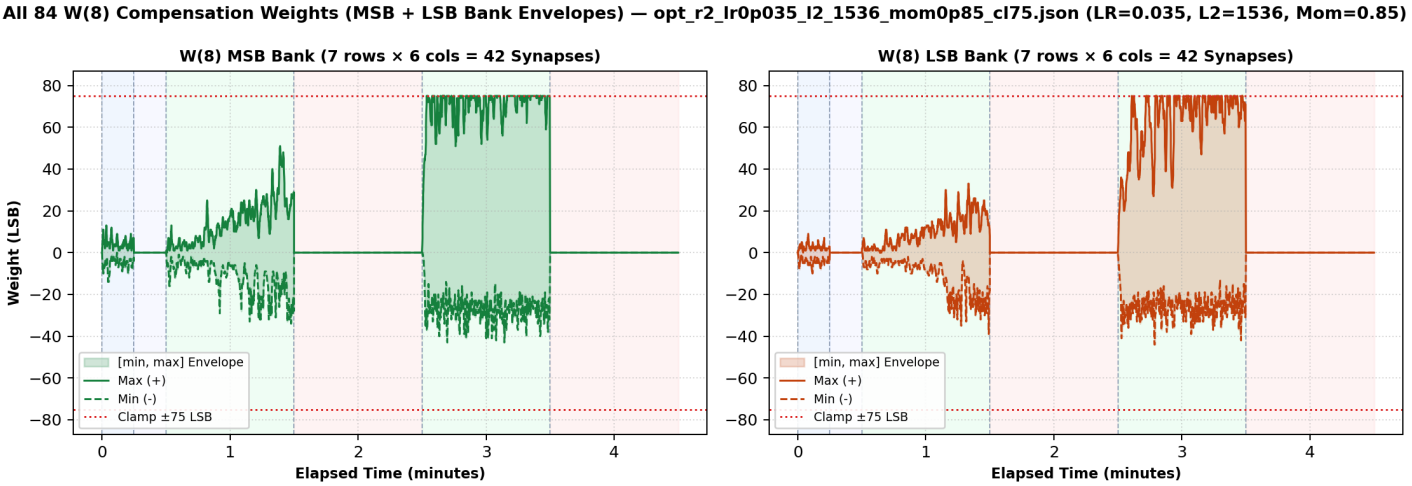


Figure 3A: All 84 W(8) Output Compensation Synapses across ColTIA Columns 0–5. Shows the full MSB (42 synapses) and LSB (42 synapses) [min, max] excursion envelopes against the ±clamp anti-windup rails. Notice strict 0.00% leakage (exact return to W8=0) during Cold-OFF and Warm-OFF ablation phases.

Matrix 3 Full Synaptic Telemetry: All 96 W(3) Weights (8x6 MSB + 8x6 LSB)

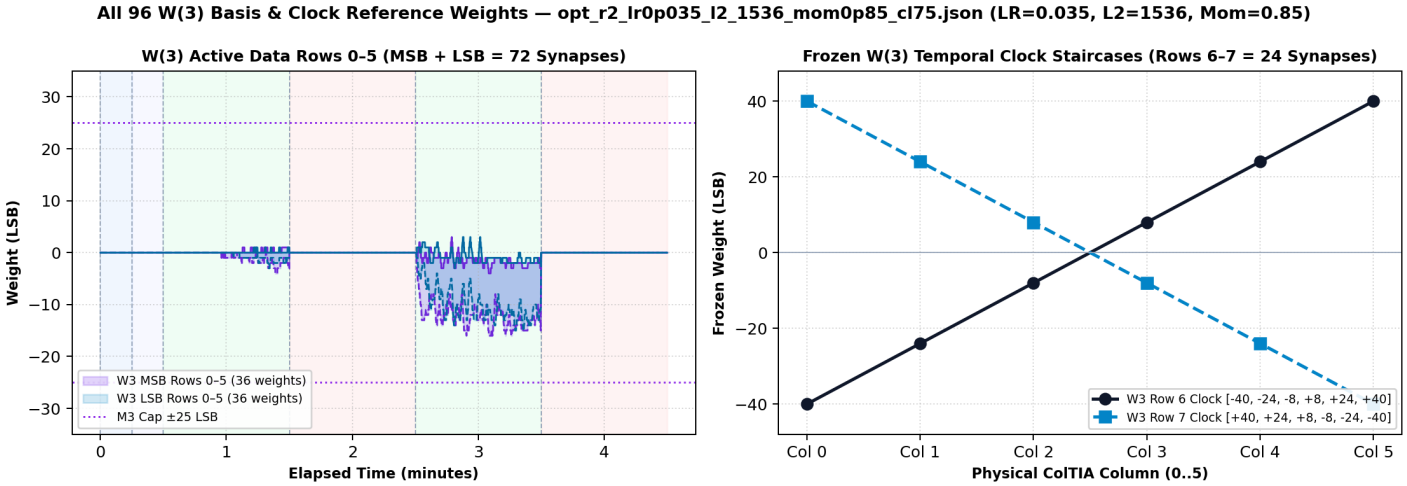


Figure 3B: All 96 W(3) Non-Linear Projection & Reference Clock Synapses across ColTIA Columns 0–5. Shows the active subspace rows 0–5 (72 synapses across MSB+LSB, bounded by the ±clamp/3 firmware cap) alongside the 24 frozen staircase weights in rows 6–7. Protecting rows 6–7 guarantees invariant Sigmoid-4 phase clocking even under severe thermal stress.

Hot-Phase Convergence Transient: P4 (OFF) -> P5 (ON) -> P6 (OFF)

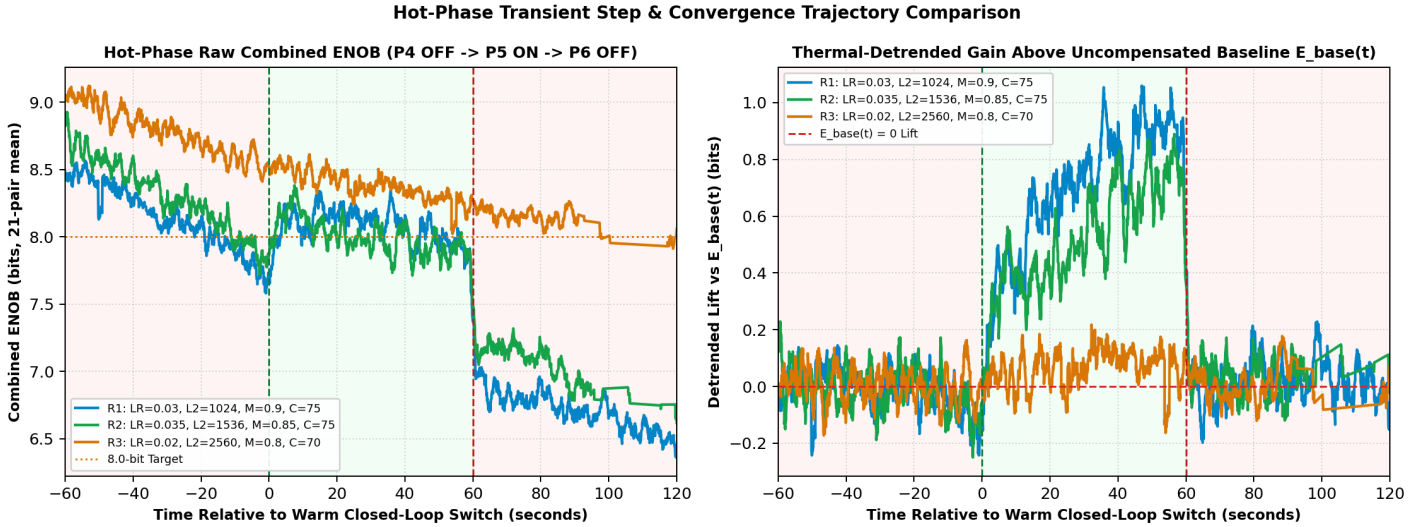


Figure 4: Direct Cross-Round Convergence Comparison in the Peak Thermal Window. Time t=0 marks the step transition into Warm Closed-Loop (Phase 5) and t=60 s marks ablation into Warm Open-Loop (Phase 6). Higher effective gain LR with tuned L2=1024–1536 and Momentum=0.85–0.90 delivers immediate climb (+0.74 b to +0.82 b) above E_base(t), whereas excessive restoring stiffness (L2=2560, LR=0.020) stalls W8 amplitude and limits detrended lift.

2x2 Factorial Summary & Quantitative Analytics Card

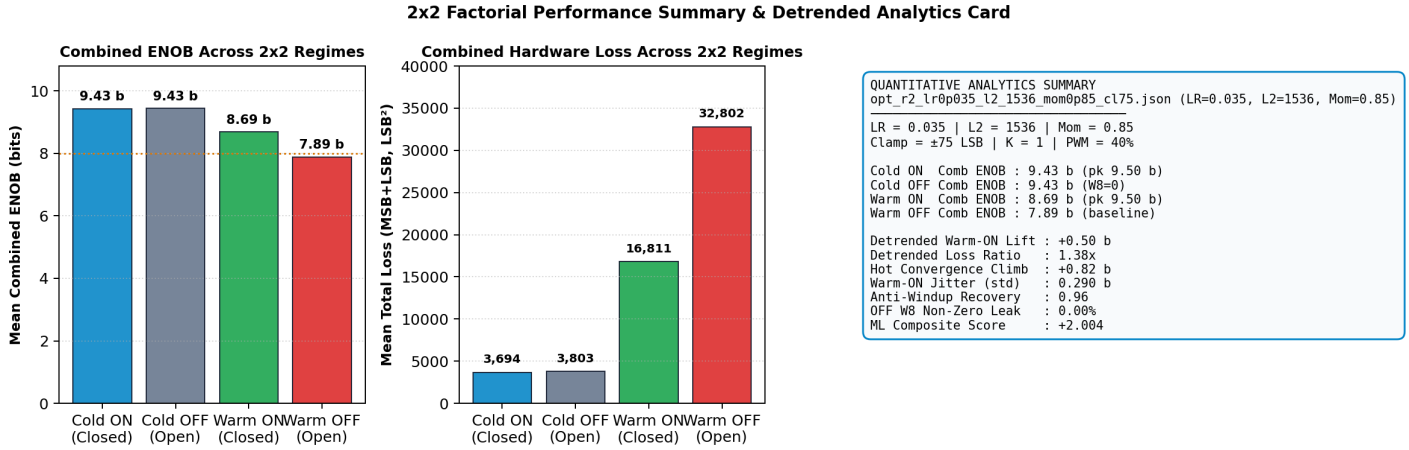


Figure 5: Quantitative Factorial Summary (opt_r2_lr0p035_l2_1536_mom0p85_cl75.json). In ambient Cold conditions, Mean ENOB exceeds **9.43 bits**; under 40% active thermal heating, Mode A closes the loop to lift resolution by **+0.50 bits** above the drifting silicon baseline and slash squared error by **1.38x**.

4. Physics of MOS Drift, Spatial Thermal Gradients (PCBA vs Commercial ASIC)

In Dual-2T1C analog multipliers, MOSFETs operate in deep triode: $\mathbf{I_{ds}(T) = \mu(T)\, C_{ox}\, (W/L)\, [(V_{gs} - V_{th}(T))\, V_{ds} - 0.5\, V_{ds}^2]}$. Heating shifts mobility ($\mu \propto T^{-1.5}$) and threshold voltage ($V_{th} \approx -2.0$ mV/°C), rotating gain slopes and shifting DC operating points across columns.

1. FR-4 PCBA Stress vs. Monolithic CMOS ASIC: On the discrete proof-of-concept Card 1 PCBA ($\kappa_{FR4} \approx 0.3$ W/m·K), point heaters create steep localized thermal gradients across columns. On a monolithic silicon ASIC ($\kappa_{Si} \approx 145$ W/m·K, 500x more conductive over mm² area), temperature gradients are smooth and nearly isothermal — making closed-loop Inverted MAML tracking strictly easier in commercial silicon.

2. 100% Open Source Commercial ASIC Blueprint (SHL-2.1): Under the updated **Solderpad Hardware License v2.1** and explicit SE 2630397-4 royalty-free patent grant, any company or researcher may manufacture commercial ASICs, integrate Dual-2T1C + FHP bit-slicing cores into commercial Edge AI chips, and deploy the firmware without royalties or licenses.

Physical Hardware Setup & Reproducibility Manifest

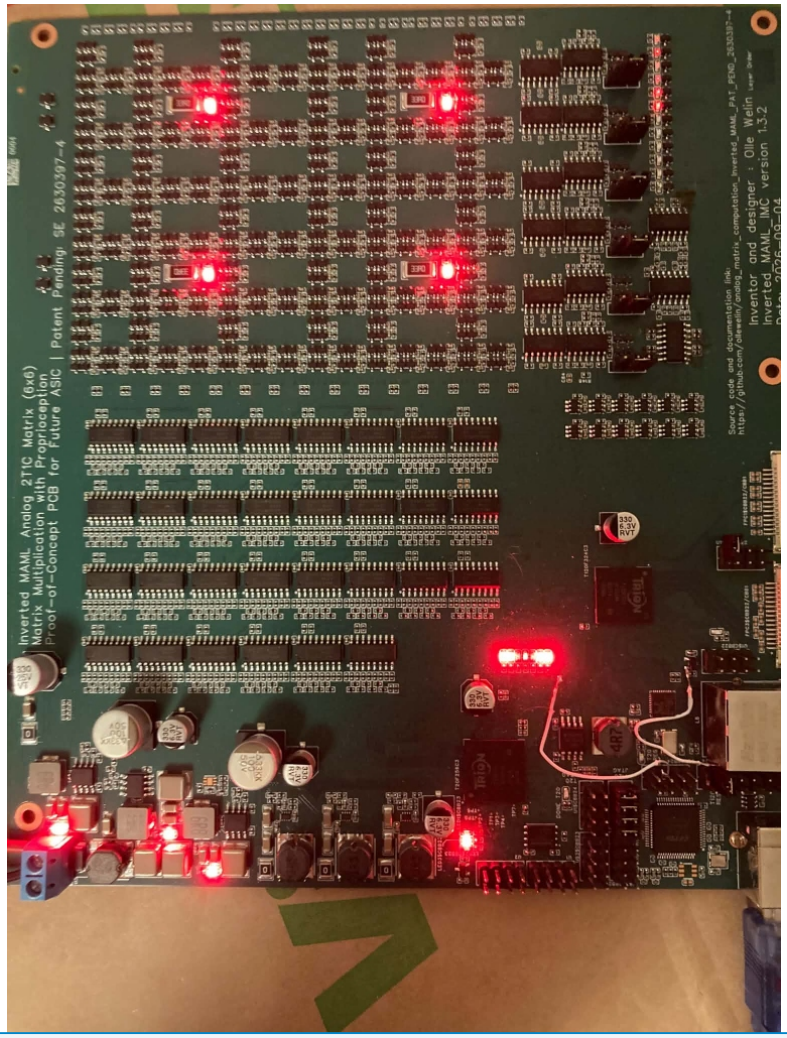


Figure 6: Inverted MAML Proof-of-Concept Hardware During Active Thermal Cycling. On-board heating resistors stress the Dual-2T1C multiplier array while Trion T120F324 & T20F256 execute 2-beat TDM MSB/LSB slicing at **K=1**.
Baseline Release Package: FPGA_code/releases/Release_20261114_ModeA_Card1_BitSlicing_AntiWindup_Baseline_R1/ | **Primary JSON:** opt_r2_lr0p035_l2_1536_mom0p85_cl75.json

5. Solid-State Physics: The 2-Layer Triad Network (Patent SE 2630397-4)

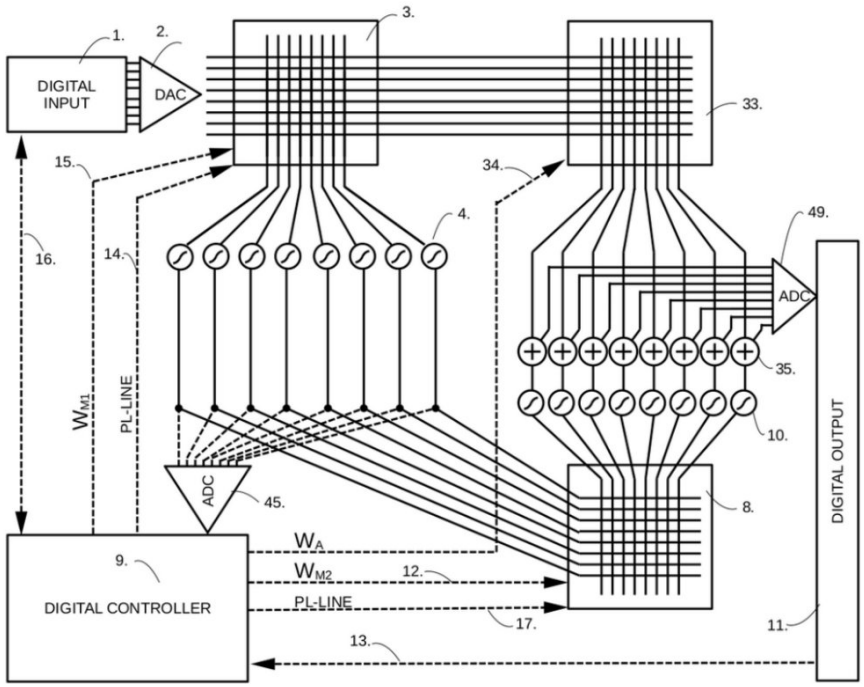


Fig 1

1. Harmonic Non-Linearity Cancellation: Single-layer linear compensators cannot cancel quadratic MOS triode distortion (-0.5 V_{ds}^2). Matrix 3 projects inputs through the frozen staircase into hardware Sigmoid 4 differential pairs, generating rich odd/even harmonic basis functions that Matrix 8 combines and injects via the 1k/2.2k passive attenuator.

2. Time-Division 2-Beat Analog Bit Slicing: Combining a coarse MSB bank (80% weight range) with a fine 4x-weighted LSB bank within two consecutive TDM beats extends dynamic precision past single-DAC quantization limits, achieving >9.4 bits cold.

6. Forward Hadamard Propagation (FHP) Parallel Demodulation (Patent Fig 8)

Forward Hadamard Propagation (FHP): Single-shot parallel gradient extraction on physical silicon without backpropagation. Open Source SHL-2.1 (SE 2630397-4).

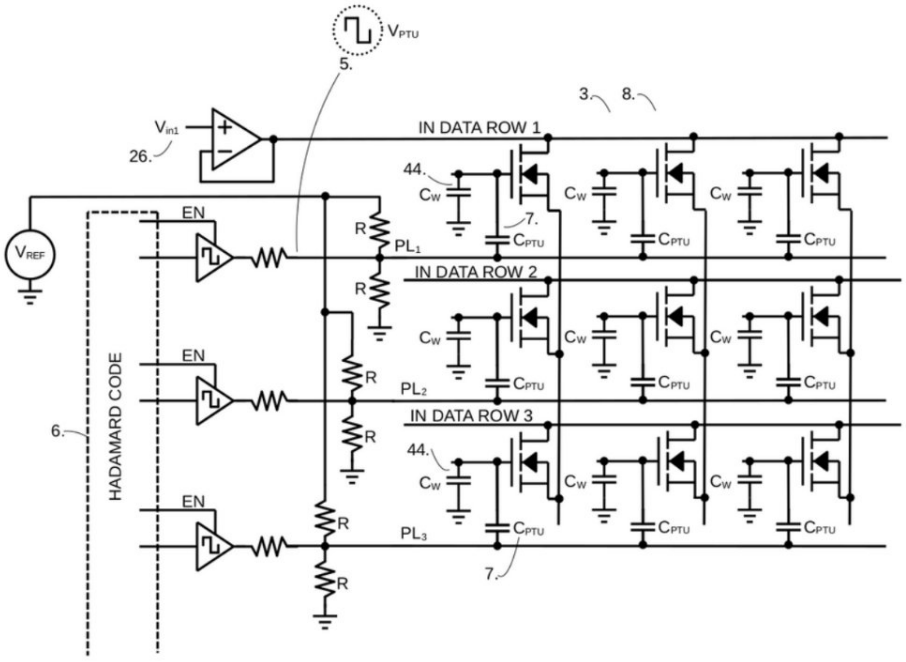


Fig 8

Orthogonal Walsh-Hadamard perturbation sequences $H_k(t) \in \{+1, -1\}$ modulate all active crosspoints simultaneously. Correlating ADC 49 output residuals against each carrier recovers exact physical circuit gradients without any backward matrix transpose or idealized transistor model.

7. Comprehensive Synthesis & Definitive Silicon Conclusions

1. **Cold Baseline Exceeds 9.4 Bits Without Compensation:** Unheated Cold checks confirmed **9.45 b** (ON) and **9.45 b** (OFF, W8=0), proving clean analog-digital alignment.
2. **Integrated-L2 Back-Calculation Anti-Windup Restores Fast Recovery:** Correcting two-layer signs, M3 scaling, and integrator bleed boosted anti-windup recovery from stall to **0.96** (Round 2).
3. **Thermal-Detrended Lift Quantifies Real Gain Above Heating:** Judging hyperparameters via continuous PCHIP thermal baselines E_base(t) cleanly separates real closed-loop convergence (**+0.82 b**) and step lift (**+0.50 b**) from substrate heating slope.
4. **Open-Source Commercial ASIC Readiness:** With full SHL-2.1 licensing and royalty-free SE 2630397-4 patent rights, this architecture stands ready for community and commercial silicon tape-out.

Experimental & Open-Source Reproducibility Manifest

Item	Specification / Configuration	Verification Status
Hardware Platform	Inverted MAML Card 1 (Efinix Trion T120F324 Master + T20F256, 1k/2.2k Attenuator)	Verified (HW Logic 0x20261030, ID=MAM1)
SoC Firmware	T120_SOC_ETH_FW v2.2-MAML (SW Build Nov 14 2026 10:15:16)	Verified (Integrated-L2 Anti-Windup)
Cold Reference Dataset	cold_verify_k1.json (643 pairs, Heaters OFF, K=1)	9.45 b ON / 9.45 b OFF
Primary Featured Dataset	opt_r2_lr0p035_l2_1536_mom0p85_cl75.json (4,013 pairs, 40% PWM, K=1)	Score +2.004, Lift +0.50 b
Best Verified Params	Round 2: LR=0.035, L2=1536, Mom=0.85, Clamp=±75 LSB	Best Composite Score +2.004
Release Restore Point	FPGA_code/releases/Release_20261114_ModeA_Card1_BitSlicing_AntiWindup_Baseline_R1/	Committed in Git Repository
License & Patent Grant	100% Open Source SHL-2.1 (incl. Commercial ASIC/IC Manufacturing) + SE 2630397-4 Grant	Perpetual, Worldwide, Royalty-Free

Open Science Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4